The Journal of Open Source Software

medRCT: Causal mediation analysis estimating interventional effects mapped to a target trial in R

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DOI: 10.xxxxxx/draft

Software

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Submitted: 25 February 2025
Published: unpublished

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Summary

Causal mediation analysis is often used to explore the causal pathways linking exposures to outcomes across various research domains, including epidemiology, public health, and social sciences. It quantifies the extent to which the causal effect of an exposure on an outcome is mediated through one or more intermediate variables, known as mediators. The goal of studies conducting mediation analyses, particularly in health research, is often to understand how hypothetical interventions targeting mediators might counter exposure effects in order to guide future treatment and policy decisions as well as intervention development.

The interventional effects (Geneletti, 2007; VanderWeele et al., 2014) are causal mediation estimands that overcome certain identifiability challenges associated with traditional estimands, known as the natural effects (Robins & Greenland, 1992). Unlike the natural effects, the interventional effects implicitly emulate target trials (Hernán & Robins, 2016) that assess the effects of distributional shifts in mediators (Moreno-Betancur & Carlin, 2018). As a result, interventional effects are more closely aligned with the goal of examining the potential impacts of hypothetical interventions that shift mediators. However, the original definitions of interventional effects do not necessarily assess shifts that are relevant to policy and practice. To bridge this gap, Moreno-Betancur et al. (2021) proposed definitions of interventional effects that map explicitly to a target trial assessing policy-relevant hypothetical mediator interventions.

medRCT provides a user-friendly interface for estimating policy-relevant interventional effects as defined in Moreno-Betancur et al. (2021) using a Monte Carlo simulation-based g-computation approach. Key features of medRCT include support for multiple mediators, intermediate confounders (exposure-induced mediator-outcome confounders), and a Shiny application (Chang et al., 2024) for comprehensive model assessment. Detailed methods and estimands implemented in medRCT are described in Moreno-Betancur et al. (2021).

Statement of need

Several software packages are available for causal mediation analysis. Existing tools, such as `mediation` (Tingley et al., 2014) and `medflex` (Steen et al., 2017), focus primarily on natural effects estimands, which are prone to identification challenges and rely on assumptions that are difficult to validate in real-world studies (Vansteelandt & Daniel, 2017). While software like `medoutcon` (Hejazi et al., 2022) leverages causal machine learning methods to estimate interventional effects, they do not consider estimands defined explicitly in terms of a target trial examining policy-relevant interventions. This limits their direct practical relevance in informing decision-making.

The medRCT package addresses these gaps by estimating policy-relevant interventional effects,

explicitly mapped to a target trial. There are several published applications of this methodology in real-world studies, including Dashti et al. (2022), Goldfeld et al. (2023), and Afshar et al. (2024), demonstrating its utility across diverse research contexts. For instance, Dashti et al. (2022) studied the mediating roles of C-reactive protein, leptin, fasting insulin, and estradiol in the effect of adiposity on cancer risk among postmenopausal women. Goldfeld et al. (2023) assessed the extent to which inequities could be mitigated by improving disadvantaged children's parental mental health and preschool attendance. Before the development of `medRCT`, these studies had to rely on user-written code, which was prone to errors and limited both accessibility and reproducibility. Several ongoing studies at the Murdoch Children's Research Institute are now using `medRCT`. For example, one study aims to investigate the impact on cardiovascular outcomes of a hypothetical intervention that shifts the distribution of inflammatory markers in adolescents from high-income households to the levels in those from low-income households, using data from multiple longitudinal cohort studies.

`medRCT` has also been used in education and training. It has been used in workshops and training sessions, such as the ViCBiostat Summer School and workshop at the Society for Epidemiologic Research 2024 meeting.

The package has been extensively documented on its website. Future developments for `medRCT` include extending its functionality to handle survey design weights to broaden its applicability to real-world studies involving complex survey designs.

Acknowledgements

This work was supported by an Investigator Grant fellowships to MMB [grant ID: 2009572] and SGD [grant ID: 2027171] from the National Health and Medical Research Council. The Murdoch Children's Research Institute is supported by the Victorian Government's Operational Infrastructure Support Program.

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